TOR AMPLITUDE IN AN INTERVAL IS PRESERVED BY THE DERIVED PULLBACK ALONG A MORPHISM OF RINGED SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the choice of the flat representative, the description of the pulled-back complex, flatness of its terms, and the second application of the characterisation.

Editorial note. The label lemma-tor-amplitude-pullback names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Representation of tor amplitude by complexes of flat modules. Same chapter, the lemma characterising tor amplitude in [a, b] through a representative by a complex of flat modules vanishing outside the interval. Supplies the representative used at the start and the criterion applied at the end.
- Flatness is preserved by pullback. Modules on Sites, the lemma stating that the pullback of a flat module is flat. Supplies flatness of the terms of the pulled-back complex.

Source reference. Original reference 0654: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12432–12432. Prerequisite pointer invoked at proof line(s) 12554; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05VD: sources/stacks-chow/tags.txt, source lines 7547–7547. Prerequisite pointer invoked at proof line(s) 12559; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(f, f^\sharp) : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed sites. Let E be an object of $D(\mathcal{O}_\mathcal{D})$. If E has tor amplitude in $[a, b]$, then Lf^*E has tor amplitude in $[a, b]$.*

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Proof. Assume E has tor amplitude in $[a, b]$. By Lemma 0654 we can represent E by a complex of $\mathcal{E}^\bullet$ of flat $\mathcal{O}$-modules with $\mathcal{E}^i = 0$ for $i \notin [a, b]$. Then Lf^*E is represented by $f^*\mathcal{E}^\bullet$. By Modules on Sites, Lemma 05VD the module $f^*\mathcal{E}^i$ are flat. Thus by Lemma 0654 we conclude that Lf^*E has tor amplitude in $[a, b]$. $\square$